COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE-010: Data Structures and Algorithm	**SECTION:**	CPE21S3
DATE PERFORMED:	7/28/26	**DATE SUBMITTED:**	8/1/26
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Seatwork 4.1: Array-Based vs. STL Stack Implementations

1. PROCEDURE OUTPUT

What procedures did you perform?

ILO A:

CODE:

ILO_A > main.cpp > main()

```cpp
#include <iostream>
#include <stack>

int main(){

    std::stack<int> newStack; // Created object
    std::stack<int> swapStack; // For swapping

    // Testing the empty() function to check if stack is empty
    std::cout <<"Is the stack empty? "<< newStack.empty()<<std::endl;

    // Insert 10 elements in a for loop
    for(int i = 0; i < 10; i++){
        newStack.push(i);
        swapStack.push(i);

        // Prints out the first top
        std::cout <<"Top of the stack: "<< newStack.top()<<std::endl;
    }

    swapStack.swap(newStack);
    std::cout << "newStack new top value: " << newStack.top()<<std::endl;


    newStack.emplace(10);

    // Prints out the stack size
    std::cout <<"Size of the stack is: "<< newStack.size()<<std::endl;


    // Removing elements in a for loop
    while(!newStack.empty()){
        std::cout <<"Remaining elements of the stack is: "<< newStack.size()<<std::endl;
        newStack.pop();
    }
    std::cout << "No more elements remaining \n";
}
```

OUTPUT:

```
PS C:\Users\TIPQC\Documents\SANTOS\HOA4p1\ILO_A> cd "c:\Users\TIPQC\Documents\SANTOS\HOA4p1\ILO_A\" ; if ($?) { g++ main.cpp -o main } ; if ($?) { .\main }
Is the stack empty? 1
Top of the stack: 0
Top of the stack: 1
Top of the stack: 2
Top of the stack: 3
Top of the stack: 4
Top of the stack: 5
Top of the stack: 6
Top of the stack: 7
Top of the stack: 8
Top of the stack: 9
newStack new top value: 9
Size of the stack is: 11
Remaining elements of the stack is: 11
Remaining elements of the stack is: 10
Remaining elements of the stack is: 9
Remaining elements of the stack is: 8
Remaining elements of the stack is: 7
Remaining elements of the stack is: 6
Remaining elements of the stack is: 5
Remaining elements of the stack is: 4
Remaining elements of the stack is: 3
Remaining elements of the stack is: 2
Remaining elements of the stack is: 1
No more elements remaining
PS C:\Users\TIPQC\Documents\SANTOS\HOA4p1\ILO_A>
```

Code Analysis: I included a library for stack to access the commands to make stacks. I then declared to stacks with “newStack” being my main stack and “swapStack” for testing out the swapping commands. The empty command returns a boolean value to tell whether the stack is empty or not. Then I used a for loop to push 10 elements inside my main stack to insert 10 different values along with the other stack. Then I display the current top value of the main stack using the top command. Then I swapped the values of both stacks and used an emplace command to add a value of 10 at the top of the main stack. Afterwards, I checked the size of the stack and also deleted all the values of the main stack by using the pop command.

ILO B.1:

CODE:

ILO_B > main.cpp > displayAll()

```cpp
#include <iostream>

const size_t maxCap = 100;
int stack[maxCap];
int top = -1, newData, elements;

// Prototyping
void push();
void pop();
void Top();
bool isFull();
bool isEmpty();
void displayAll();

int main(){
    int choice;

    std:: cout << "Enter number of max element for new stack (max: 100)";
    std::cin >> elements;

    while(true){
        std::cout << "Stack operations: "<<std::endl;
        std::cout << "1. Push \n2. Pop \n3. Top \n4. isEmpty \n5. isFull \n6. Display All \n";
        std::cin >> choice;

        switch(choice){
            case 1: push();
            break;

            case 2: pop();
            break;

            case 3: Top();
            break;

            case 4: std::cout << isEmpty() << std::endl;
            break;

            case 5: std:: cout << isFull() << std::endl;
            break;

            case 6: displayAll();
            break;

            default: std::cout << "Invalid Choice" << std::endl;
            break;
        }
    }


```

```
    return 0;
}

bool isEmpty(){
    // Top is -1
    if(top == -1) return true;
    return false;
}

bool isFull(){
    // != maxCap
    if(top >= elements){
        std::cout << "Stack Overflow";
        return true;
    }
return false;
}

void push(){

    // Check if stack is full, if yes return error
    if(isFull())return;

    // Ask user for value to be pushed in stack
    std::cout<<"What number do you want to add to the stack? ";
    std::cin>> newData;

    // Store inside stack
    stack[++top] = newData;

}

void pop(){

    // Check if there are no more elements inside the stack
    if(isEmpty()) return;

    // Display element to be popped:
    std::cout<<"Popping: "<< stack[top];

    // Decrement top
    --top;
}

void Top(){
    // Error check
    if(isEmpty()) return;

    // Display top element
    std::cout << "The top element is: " << stack[top] << std::endl;
```

```
101 }
102
103 void displayAll(){
104     // Error check
105     if(isEmpty()) return;
106
107     std::cout << "Values inside the stack: \n";
108     for(int i = 0; i < elements; i++){
109         std::cout << stack[i] << std::endl;
110     }
111
112 }
113
```

OUTPUT:

```
PS C:\Users\TIPQC\Documents\SANTOS\HOA4p1\ILO_B> cd "c:\Users\TIPQC\Documents\SANTOS\HOA4p1\ILO_B\" ; if ($?) { g++ main.cpp -o main } ; if ($?) { .\main }
Enter number of max element for new stack (max: 100)100
Stack operations:
1. Push
2. Pop
3. Top
4. isEmpty
5. isFull
6. Display All
1
What number do you want to add to the stack? 12
Stack operations:
1. Push
2. Pop
3. Top
4. isEmpty
5. isFull
6. Display All
1
What number do you want to add to the stack? 23
Stack operations:
1. Push
2. Pop
3. Top
4. isEmpty
5. isFull
6. Display All
1
What number do you want to add to the stack? 34
Stack operations:
1. Push
2. Pop
3. Top
4. isEmpty
5. isFull
6. Display All
2
Popping: 34Stack operations:
1. Push
2. Pop
3. Top
4. isEmpty
5. isFull
6. Display All
3
The top element is: 23
Stack operations:
1. Push
2. Pop
3. Top
4. isEmpty
5. isFull
6. Display All
```

```
Popping: 34Stack operatic
1. Push
2. Pop
3. Top
4. isEmpty
5. isFull
6. Display All
3
The top element is: 23
Stack operations:
1. Push
2. Pop
3. Top
4. isEmpty
5. isFull
6. Display All
4
0
Stack operations:
1. Push
2. Pop
3. Top
4. isEmpty
5. isFull
6. Display All
5
0
Stack operations:
1. Push
2. Pop
3. Top
4. isEmpty
5. isFull
6. Display All
6
Values inside the stack:
12
23
34
0
0
0
0
0
0
0
0
0
0
0
0
0
0
```

Code Analysis: I declared a constant size of allowed bytes as the max capacity of 100 to limit the stack elements allowed along with three integers to use for later. I prototyped six functions and called the int main. I then made a menu for the command line interface for the functions. Using switch case, I called every function per case and a default as an invalid choice.

ILO_B.2:

main.cpp

ILO_B2 > main.cpp > main()

```cpp
#include <iostream>
#include "listStack.h"

int main(){
    std::cout << "------- Pushing 'I' 'E' 'E' 'E' -------\n";

    //Push
    push('I');
    push('E');
    push('E');
    push('E');

    std::cout << "Display All: ";
    displayAll<char>();

    // Display top
    Top<char>();

    // Pop and print
    std::cout << "------- Popping 'E' 'E' 'E' 'E' -------\n";
    while(isEmpty<char>()){
        std::cout<<"Popping: " << pop<char>() << std::endl;
    }

    //Test stack underflow
    pop<char>();

    return 0;
}
```

listStack.h

ILO_B2 > C listStack.h > displayAll<T>()

```cpp
#ifndef LIST_STACK_H
#define LIST_STACK_H

// Class node with data and next members
template <typename T>
class Node{
    public:
    T data;
    Node<T> *next = nullptr;
};

template<typename T>
Node<T>* head = nullptr;
// Create pointer

template <typename T>
void push(T newData){
    // Create new node
    Node<T>* newNode = new Node<T>;

    // Assign data to new node
    newNode->data = newData;

    // Pointer the new node next to current top
    newNode->next = head<T>;

    // Assign the new top to the new node
    head<T> = newNode;
 }

template <typename T>
bool isEmpty(){
    return head<T> == nullptr;
}

template <typename T>
T pop(){
    // If stack is empty, error catch
    if(isEmpty<T>()){
        std::cout << "Stack Underflow \n";
    }

    // Saving current top node
    Node<T> *temp = head<T>;

    // Store data to temp variable
    T tempVal = temp->data;

    // Move head to next node
    head<T> = head<T>->next;

    // Deallocate and free memory
    delete temp;

    // Return temporary variable
    return tempVal;

}
```

```
template<typename T>
void Top(){
    if(isEmpty<T>())return;
    std:: cout << "The top is: " << head<T>->data;
}

template <typename T>
void displayAll() {
    if (isEmpty<T>()) return;

    while (head<T> != nullptr) {
        std::cout << head<T>->data << " ";
        head<T> = head<T>->next;
    }
    std::cout << std::endl;
}

#endif
```

OUTPUT:

```
------- Pushing 'I' 'E' 'E' 'E' -------
Display All: E E E I
------- Popping 'E' 'E' 'E' 'E' -------
Popping: Stack Underflow
PS C:\Users\TIPQC\Documents\SANTOS\HOA4p1\ILO_B2>
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

ILO C - 1:
CODE:

Supplementary_A > main.cpp > Array

```cpp
#include <iostream>
#include <string>

// Declare class for array
class Array {
private:
    char arr[100]; // Max
    int top;

public:
    Array() {
        top = -1; // Array -> Stack
    }

    bool isEmpty() {
        return top == -1;
    }

    bool isFull() {
        return top == 100 - 1;
    }

    void push(char ch) {
        if (!isFull()) {
            arr[++top] = ch; // Pre increment to change top
        }
    }

    char pop() {
        if (!isEmpty()) {
            return arr[top--]; // Post decrement for array return
        }
        return '\0'; // Null character value
    }
};

// Pairing and checking
bool pairSymbols(char open, char close) {
    return (open == '(' && close == ')') || (open == '{' && close == '}') || (open == '[' && close == ']');
}


bool balSymbols(std::string expr) {
    Array stack;

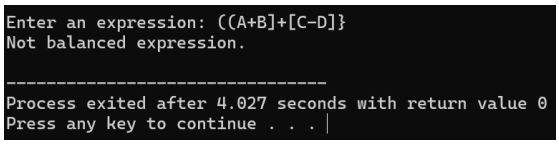
    // for loop for checking expressions, then push the character
    for (char ch : expr) {
        if (ch == '(' || ch == '{' || ch == '[') {
            stack.push(ch);
        }

        // For closing symbols, push it on top and check for conditions:
        else if (ch == ')' || ch == '}' || ch == ']') {
            if (stack.isEmpty())
                return false;

            // Put closing symbol on top
            char top = stack.pop();
```

```
60              // If not matching opening, reject
61              if (!pairSymbols(top, ch))
62                  return false;
63          }
64      }
65      return stack.isEmpty();
66  }
67
68  int main() {
69      // Call expression in string library
70      std::string expression;
71      std::cout << "Enter an expression: ";
72
73      // Output and input here
74      getline(std::cin, expression);
75
76      // Check for balancing
77      if (balSymbols(expression))
78          std::cout << "Balanced expression." << std::endl;
79      else
80          std::cout << "Not balanced expression." << std::endl;
81
82      return 0;
83  }
```

Expression	Valid? (Y/N)	Output (Console Screenshot)	Analysis
(A+B)+(C-D)	Y	Enter an expression: (A+B)+(C-D) Balanced expression. -------------------------------- Process exited after 42.9 seconds with return value 0 Press any key to continue . . .	Symbols are balanced due to having two pairs of parenthesis
((A+B)+(C-D)	N	Enter an expression: ((A+B)+(C-D) Not balanced expression. -------------------------------- Process exited after 6.133 seconds with return value 0 Press any key to continue . . .	Symbols are balanced due to having two pairs of parenthesis
((A+B)+[C-D])	Y	Enter an expression: ((A+B)+[C-D]) Balanced expression. -------------------------------- Process exited after 6.005 seconds with return value 0 Press any key to continue . . .	Symbols are balanced due to having a pair of parenthesis and a pair of brackets inside another pair of

			parenthesis.
((A+B]+[C-D]}	**N**		Symbols are not balanced due to having a missing pair of a bracket, brace and parenthesis.

ILO C - 2:
CODE:

```
#include <iostream>
#include <string>

class Node {
public:
    char data;
    Node* next; // Node points to next

    Node(char x) {
        data = x; // Storing char
        next = NULL; // Pointer set to null
    }
};

class StackLL { // Stack commands remaking
private:
    Node* top; // Will always point to top

public:
    StackLL() {
        top = NULL; // Set to null to be an empty stack at first
    }

    bool isEmpty() {
        return top == NULL;
    }

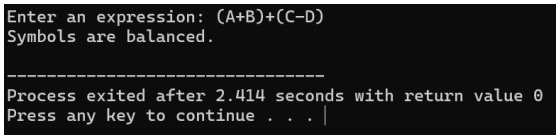
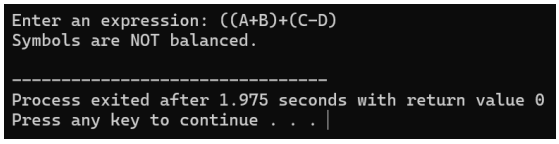
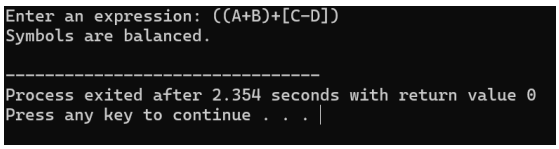
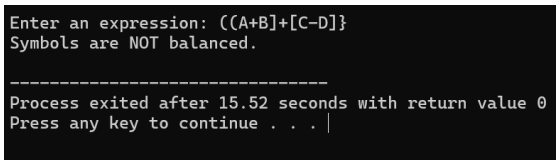
    void push(char ch) {
        Node* newNode = new Node(ch);
        newNode->next = top;
        top = newNode;
    }

    char pop() {
        if (isEmpty()) return '\0'; // Null char

        // Using temp to do replicate pop command
        Node* temp = top;
        char value = temp->data;
        top = top->next;
```

```
41          delete temp; return value;
42      }
43  };
44
45  // Prototyping
46  bool pairSymbols(char open, char close);
47  bool balSymbols(std::string expr);
48
49  int main() {
50
51      // Declare string expr for input
52      std::string expr;
53      std::cout << "Enter an expression: ";
54
55      // Output and input here
56      std::getline(std::cin, expr);
57
58      // Check for balancing
59      if (balSymbols(expr))
60          std::cout << "Symbols are balanced." << std::endl;
61      else
62          std::cout << "Symbols are NOT balanced." << std::endl;
63
64      return 0;
65  }
66
67  // Pairing and checking
68  bool pairSymbols(char open, char close) {
69      return (open == '(' && close == ')') || (open == '{' && close == '}') || (open == '[' && close == ']');
70  }
71
72  bool balSymbols(std::string expr) {
73      StackLL s;
74      // for loop for checking expressions, then push the character
75      for (int i = 0; i < expr.length(); i++) {
76          char ch = expr[i];
77          if (ch == '(' || ch == '{' || ch == '[') {
78              s.push(ch);
79          }
80
81          // For closing symbols, push it on top and check for conditions:
82          else if (ch == ')' || ch == '}' || ch == ']') {
83              if (s.isEmpty()) return false;
84
85              // Put closing symbol on top
86              char top = s.pop();
87
88              // If not matching opening, reject
89              if (!pairSymbols(top, ch)) return false;
90          }
91      }
92      return s.isEmpty();
93  }
```

Expression	Valid? (Y/N)	Output (Console Screenshot)	Analysis

(A+B)+(C-D)	**Y**		Symbols are balanced due to having two pairs of parenthesis
((A+B)+(C-D)	**N**		Symbols are not balanced due to having one extra open parenthesis.
((A+B)+[C-D])	**Y**		Symbols are balanced due to having a pair of parenthesis and a pair of brackets inside another pair of parenthesis.
((A+B]+[C-D]}	**N**		Symbols are not balanced due to having a missing pair of a bracket, brace and parenthesis.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or

completion of the required work.

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